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27-02-2025 / I

JECRC UNIVERSITY

School of Engineering

I- In Term Examination, Feb 2025

B. Tech. - II Semester

Subject: Engineering Mathematics II (DMA002A)

Max. Marks: 40

Time: 1¼ hrs.

Instructions:

1. All questions are compulsory.
2. Do not write anything other than your Reg. No. on the question paper.
3. Programmable calculators are not permitted.
4. Write the question numbers clearly while answering.

CO1: Use matrices, determinants and techniques for solving systems of linear equations in the different areas of Linear Algebra. Understand the definitions of Vector Space and its linear independence. Solve Eigen value problems and apply Cayley Hamilton Theorem.

CO2: Understanding convergence of sequence and series.

Part -A

(2x2=4)

A1 [CO1] Give the Statement of Cayley -Hamilton Theorem.

A2 [CO2] Define sequence and types of sequences.

Part -B

(2x7=14)

B1 [CO1] Examine whether the system of following equations is consistent and if consistent, find its solution

$$x + 2y + z = 2, \quad 3x + y - 2z = 1, \quad 4x - 3y - z = 3, \quad 2x + 4y + 2z = 4.$$

B2 [CO2] Test the convergence of the series $\sum_{n=1}^{\infty} \frac{(2n-1)}{n(n+1)(n+2)}$.

Part -C

(2x11=22)

C1 [CO1] Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.

C2 [CO2] Test for convergence of the series, $1 + \frac{3}{7}x + \frac{3 \cdot 6}{7 \cdot 10}x^2 + \frac{3 \cdot 6 \cdot 9}{7 \cdot 10 \cdot 13}x^3 + \dots$

23-04-2025 / I

JECRC UNIVERSITY

School of Engineering

2nd In Term Examination April 2025

B. Tech; II Semester (Common to all)

Subject: Engineering Mathematics II (DMA002A)

Max. Marks: 40

Time: 3.15 hrs.

Instructions:

1. Attempt all the questions in sequence.

2. Illustrate your answers with suitable examples and diagrams, wherever necessary.

3. Write the relevant question number before writing the answer.

CO3: Identify, analyze and subsequently solve physical situations whose behavior can be described by First order and first degree ordinary differential equations and Higher order linear differential equation with constant coefficients.

CO4: Determine solutions to second order linear differential equations with variable coefficients.

 $2 \times 2 = 4$ **Part A**

A1. [CO3] Solve: $\left(y - x \frac{dy}{dx}\right) = a \left(y^2 + \frac{dy}{dx}\right)$

A2. [CO4] Is the differential equation $(2x^2 + 3x) \frac{d^2y}{dx^2} + (6x + 3) \frac{dy}{dx} + 2y = (x + 1)e^x$, is exact.

Part B $2 \times 7 = 14$

B1. [CO3] Solve: $\frac{dy}{dx} = \frac{2x - y + 1}{x + 2y - 3}$

B2. [CO4] Solve: $\cos x \frac{d^2y}{dx^2} + \sin x \frac{dy}{dx} - 2y \cos^3 x = 2 \cos^5 x$

Part C $2 \times 11 = 22$

C1. [CO3] Solve: $(D^3 + 2D^2 + D)y = e^{-x} + \cos x + x^2$

C2. [CO4] Solve by method of variation of parameters

$$(D^2 + 4)y = 4 \tan 2x$$

JECRC UNIVERSITY

School of Engineering

III In Term Examination -2024-25

B.Tech- II Semester (Common to all)

Subject: Engineering Mathematics II (DMA402A)

Max. Marks: 40

Time: 1:25 hrs.

Instructions:

1. Attempt all the questions in sequence.

2. Illustrate your answers with suitable examples and diagrams, wherever necessary.

3. Write the relevant question number before writing the answer.

CO1: Use matrices, determinants and techniques for solving systems of linear equations in the different areas of Linear Algebra. Understand the definitions of Vector Space and its linear independence. Solve Eigen value problems and apply Cayley Hamilton Theorem.

CO2: Understanding convergence of sequence and series.

CO3: Identify, analyze and subsequently solve physical situations whose behavior can be described by First order and first degree ordinary differential equations and higher order linear differential equation with constant coefficients.

CO4: Determine solutions to second order linear differential equations with variable coefficients.

CO5: Understanding the series solutions of second order linear differential equations with variable coefficients

Part A (2X2)=4

A1.(CO1) Find the rank of matrix $\begin{bmatrix} 1 & 2 & 3 & -1 \\ 2 & 1 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$.

A2.(CO3) Solve $(4x+y)^2 \frac{dx}{dy} = 1$.

Part B (7X2)=14

B1.(CO3) Solve $(x^2 y^2 + 3y)dx + (3x^2 y - x)dy = 0$.

B2.(CO4) Solve $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = x^2 + \frac{1}{x^2}$.

Part C (11X2)=22

C1.(CO1) Find the Eigen value and Eigen vector of the matrix $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$.

C2.(CO5) Solve using Charpit's method $px + qy = pq$.